

# Castalia

A four-channel electrochemical sensing SoC with a five-hart RISC-V back end

Maxx Seminario

University of Nebraska–Lincoln

September 21, 2026

# Where cancers are caught, and whether anyone looks

Castalia

Maxx Seminario

Target application

The chip

Analog front end

Operating modes

Analog blocks

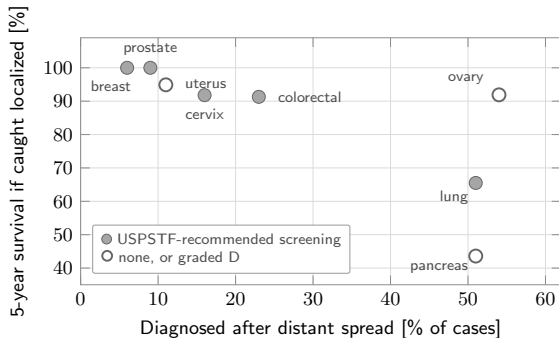
Digital

Vision

From currents to a score

References

Backup



[1] SEER 21, 2016–2022; [2, 3] USPSTF

# Why ovarian: late, lethal, and unscreened

Castalia

Maxx Seminario

Target application

The chip

Analog front end

Operating modes

Analog blocks

Digital

Vision

From currents to a score

References

Backup



| Site       | Distant at dx | Deaths/100 cases | Screening (USPSTF) |
|------------|---------------|------------------|--------------------|
| Ovary      | 54 %          | 59               | D                  |
| Pancreas   | 51 %          | 78               | D                  |
| Lung       | 51 %          | 54               | B, LDCT            |
| Colorectal | 23 %          | 35               | A                  |
| Cervix     | 16 %          | 31               | A                  |
| Uterus     | 11 %          | 21               | none               |
| Prostate   | 9 %           | 11               | C, PSA             |
| Breast     | 6 %           | 13               | B                  |

USPSTF: A/B recommended; C offered selectively; D recommended *against*; none: no statement [7].

[1] SEER, 2026 estimates; [2, 3] USPSTF

Most lethal gynecologic cancer [1]: 92 % survivable if caught localized, yet 54 % present after spread, and screening is graded D [2]. UKCTOCS moved stage, not mortality [4].

# One serum aliquot, four sites, one score

Castalia

Maxx Seminario

Target application

The chip

Analog front end

Operating modes

Analog blocks

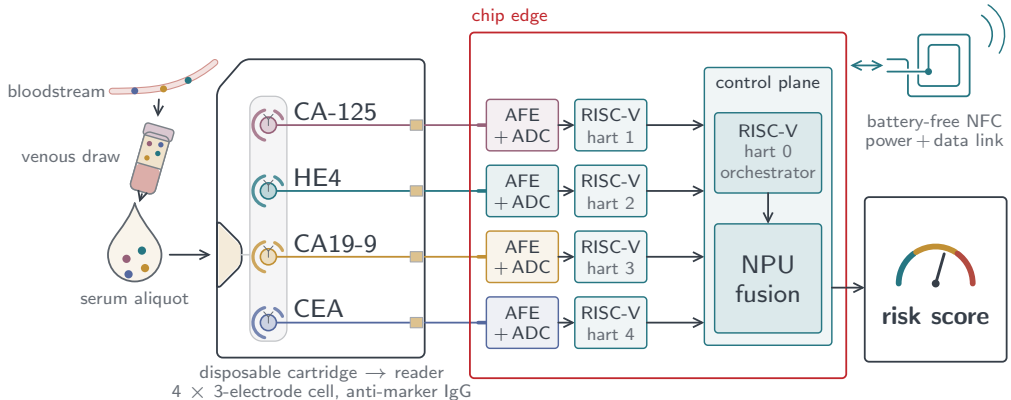
Digital

Vision

From currents to a score

References

Backup



**Symptomatic-patient triage on a disposable four-site cartridge; four markers read at the same instant, fused on chip.**

# No single marker is a test; four fused are [5]

Castalia

Maxx Seminario

Target  
application

The chip

Analog front end

Operating modes

Analog blocks

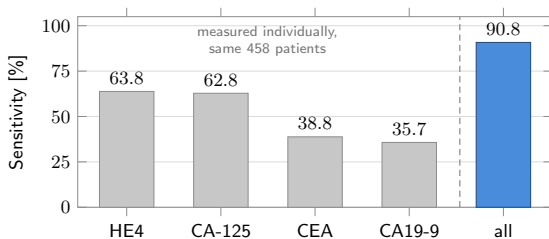
Digital

Vision

From currents to  
a score

References

Backup



| Marker | Molecule                          | Cut-off   |
|--------|-----------------------------------|-----------|
| CA-125 | MUC16 mucin epitope [8]           | 35 U/mL   |
| HE4    | WFDC2 glycoprotein [6]            | 70 pmol/L |
| CA19-9 | sialyl-Lewis <sup>a</sup> antigen | 37 U/mL   |
| CEA    | CEACAM5 glycoprotein              | 5 ng/mL   |

Each marker covers a subtype the others miss. The decision is a learned combination, so the chip is four channels plus the processor that fuses them.

# Block diagram

Castalia

Maxx Seminario

Target application

The chip

Analog front end

Operating modes

Analog blocks

Digital

Vision

From currents to a score

References

Backup



Four channel tiles own four AFE sites; every master reaches ROM, RAM, NPU and peripherals only through the arbiter.

# Physical implementation

Castalia

Maxx Seminario

Target  
application

The chip

Analog front end

Operating modes

Analog blocks

Digital

Vision

From currents to  
a score

References

Backup

**ARTWORK PENDING**



**ARTWORK  
PENDING**



# Grounded-WE two-amplifier potentiostat

Castalia

Maxx Seminario

Target application

The chip

Analog front end

Operating modes

Analog blocks

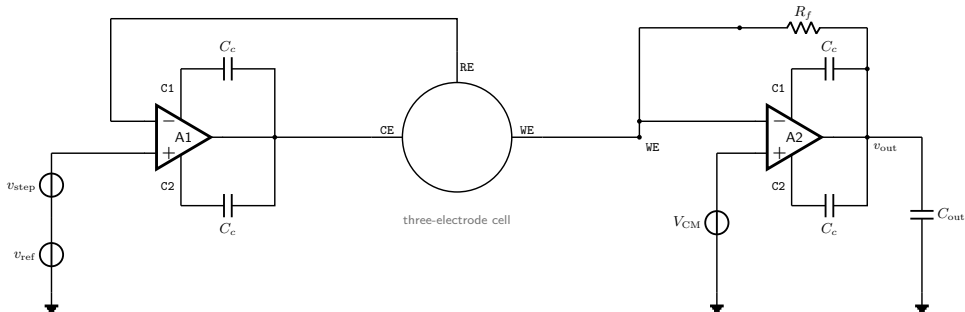
Digital

Vision

From currents to a score

References

Backup



**A1 drives CE with RE in feedback; A2 holds WE at  $v_{cm}$  through  $R_f$ .  $v_{out} = v_{cm} - I_{we}R_f$ , both polarities.**

# Analog top level

Castalia

Maxx Seminario

Target  
application

The chip

Analog front end

Operating modes

Analog blocks

Digital

Vision

From currents to  
a score

References

Backup

**ARTWORK PENDING**



Virtuoso export of the analog top level: four channels, shared bias, test bus

# The electrode model, in Verilog-A

Castalia

Maxx Seminario

Target application

The chip

Analog front end

Operating modes

Analog blocks

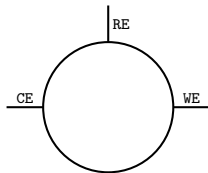
Digital

Vision

From currents to a score

References

Backup



$R_u$ ,  $R_{ce}$ ,  $C_{dl}$  lumped; the faradaic branch  $Z_f$  is the Verilog-A element.

$i_f$  faradaic current  
 $\eta$  overpotential,  $E - E^{0'}$   
 $f$   $F/RT$   
 $C_{O,R}(0, t)$  surface conc. of O, R  
 $C^*$  bulk conc.  
 $\delta$  Nernst layer

Kinetics  $i_f = nFAk^0 [C_R(0, t) e^{(1-\alpha)nf\eta} - C_O(0, t) e^{-\alpha nf\eta}]$   
 Transport  $\partial C / \partial t = D \partial^2 C / \partial x^2$ , finite-volume, 28 expanding slabs per species  
 Boundaries flux =  $\pm i_f / nFA$  at the surface;  $C(\delta, t) = C^*$  at the Nernst layer

| Parameter | Symbol   | Default             | Unit                        | Quantity                             |
|-----------|----------|---------------------|-----------------------------|--------------------------------------|
| E0        | $E^{0'}$ | 0.0                 | V                           | formal potential of the couple       |
| nel       | $n$      | 1                   | —                           | electrons transferred                |
| alpha     | $\alpha$ | 0.5                 | —                           | transfer coefficient                 |
| k0        | $k^0$    | 0.1                 | $\text{cm s}^{-1}$          | standard heterogeneous rate constant |
| Dif       | $D$      | $5 \times 10^{-6}$  | $\text{cm}^2 \text{s}^{-1}$ | diffusion coefficient, both species  |
| Area      | $A$      | 0.01                | $\text{cm}^2$               | electrode area                       |
| Cox       | $C_O^*$  | $1 \times 10^{-6}$  | $\text{mol cm}^{-3}$        | bulk concentration, oxidised species |
| Cred      | $C_R^*$  | 0                   | $\text{mol cm}^{-3}$        | bulk concentration, reduced species  |
| Temp      | $T$      | 298.15              | K                           | interface temperature                |
| expmax    | —        | 40                  | —                           | exponent clamp on the kinetic terms  |
| gpar      | —        | $1 \times 10^{-15}$ | S                           | DC shunt across the terminals        |

Equations: Bard–Faulkner [13]. Discretisation: Britz–Strutwolf, Feldberg [17, 18]. Precedent for Verilog-A electrode co-simulation: Jog et al. [19].

# Emulating the four markers

Castalia

Maxx Seminario

Target application

The chip

Analog front-end

Operating modes

Analog blocks

Digital

Vision

From currents to a score

References

Backup

| Marker | Reporter       | $E^{0'}$ (V) | Peak (nA) | $C_R^*$ ( $\mu$ M) | Cut-off         |
|--------|----------------|--------------|-----------|--------------------|-----------------|
| CA-125 | methylene blue | -0.15        | 856       | 279                | 35 U/mL [5, 8]  |
| HE4    | ferrocene      | +0.10        | 1311      | 394                | 70 pmol/L [6]   |
| CA19-9 | TMB / HRP [20] | +0.25        | 1659      | 458                | 37 U/mL [5]     |
| CEA    | anthraquinone  | +0.40        | 939       | 303                | 5 ng/mL [5, 14] |

$E^{0'}$  reporter formal potential, i.e. peak position; chosen in the  $-0.6$  to  $+0.4$  V window [22, 23], 150–250 mV apart, not measured per reporter

Peak inside 50 nA to 5  $\mu$ A on screen-printed electrodes [14, 15, 21];  $C_R^*$  fitted to reach it

$k^0$  0.1 cm/s, reversible; nothing published on carbon for these markers

Shared  $\alpha = 0.5$ ,  $n = 1$ ,  $D = 5 \times 10^{-6}$  cm<sup>2</sup>/s,  $A = 0.01$  cm<sup>2</sup>, 37 °C,  $R_f = 560$  k $\Omega$

**Representative values, not marker-measured: these slides demonstrate the channel, not an assay.**

# Channel range, assay range, and where the panel was placed

Castalia

Maxx Seminario

Target application

The chip

Analog front end

Operating modes

Analog blocks

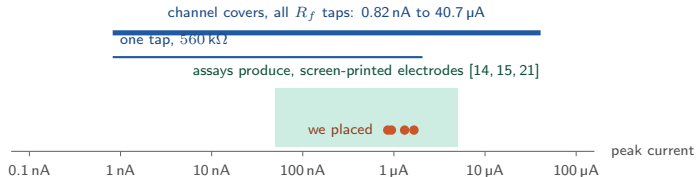
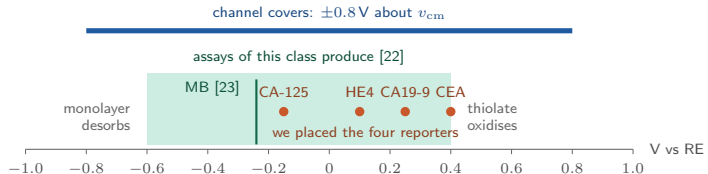
Digital

Vision

From currents to a score

References

Backup



Potentials: spaced 150–250 mV for one-sweep resolution inside the reporter window, not per-reporter values. Peak heights: inside the published envelope and one tap's full scale, then  $C_R^*$  fitted. Silicon is unaffected: the channel covers the assay band with 0.2–0.4 V and a decade of current to spare, so moving the dots changes what the simulation shows, not what the chip can measure.

# Exercising the cell: differential-pulse voltammetry

Castalia

Maxx Seminario

Target application

The chip

Analog front end

Operating modes

Analog blocks

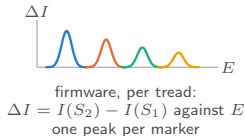
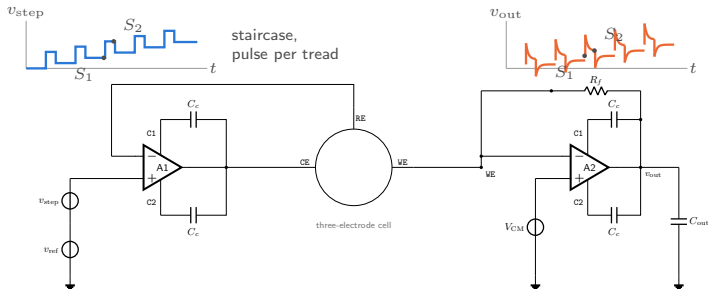
Digital

Vision

From currents to a score

References

Backup



Excite

DAC: staircase with a pulse on every tread, applied at RE by A1

Sense

$$v_{out} = v_{cm} - i_{WE} R_f$$

Sample

SAR at  $S_1$  (before the pulse) and  $S_2$  (end of the pulse), once charging has decayed

Resolve

firmware:  $\Delta I = I(S_2) - I(S_1)$  vs  $E$ ; peak position: marker, peak height: concentration

# Four markers resolved in one differential-pulse sweep

Castalia

Maxx Seminario

Target application

The chip

Analog front end

Operating modes

Analog blocks

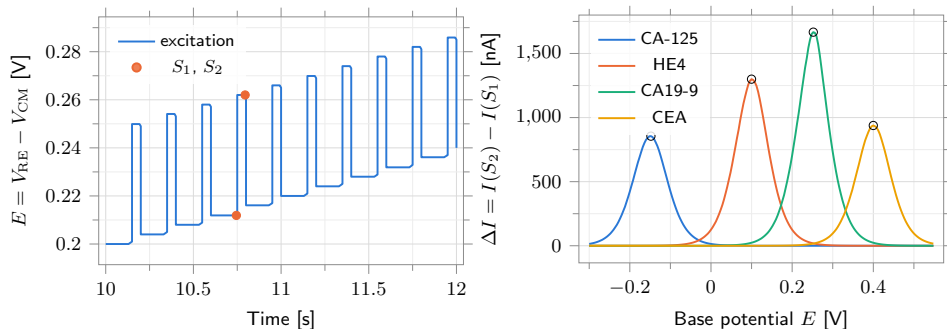
Digital

Vision

From currents to a score

References

Backup



Peak position identifies the reporter, peak height carries concentration. Apexes within 2 mV and 1 % of the assay targets; blank below 1 pA.

# Every clinical cut-off lands inside the channel window

Castalia

Maxx Seminario

Target application

The chip

Analog front end

Operating modes

Analog blocks

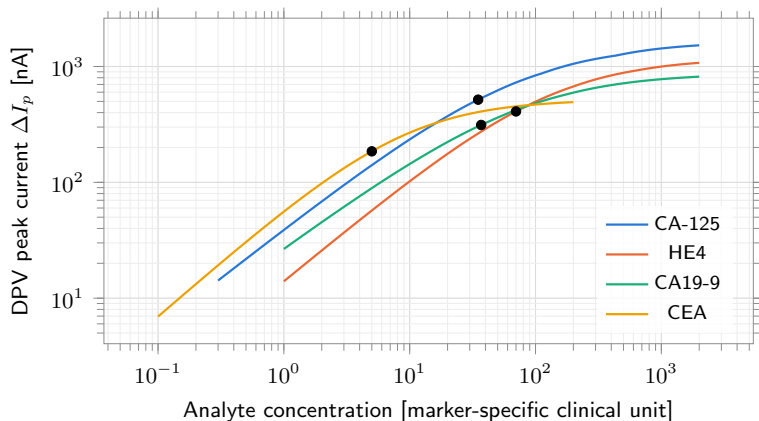
Digital

Vision

From currents to a score

References

Backup



CA-125 35 U/mL, HE4 70 pmol/L, CA19-9 37 U/mL, CEA 5 ng/mL: all four on the steep part of the curve at one  $R_f$  setting.

# Cyclic voltammetry through the assembled channel

Castalia

Maxx Seminario

Target  
application

The chip

Analog front end

Operating modes

Analog blocks

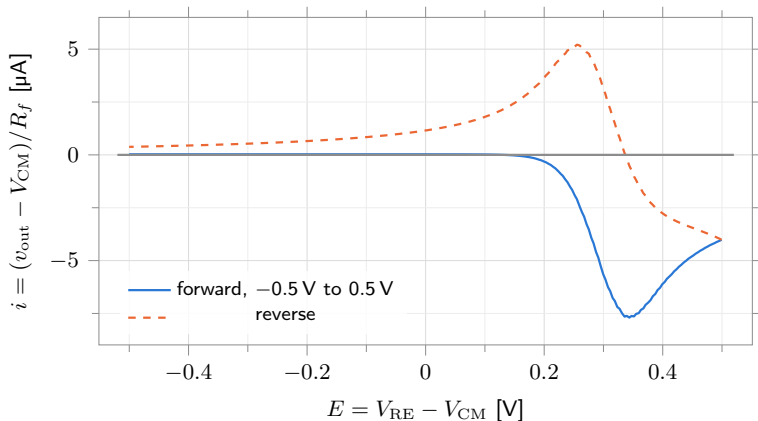
Digital

Vision

From currents to  
a score

References

Backup



Staircase potential from the on-chip DAC, current digitised by the on-chip SAR, unclipped over the sweep.

# Why impedance: for electrodes that stay in

Castalia

Maxx Seminario

Target application

The chip

Analog front end

Operating modes

Analog blocks

Digital

Vision

From currents to a score

References

Backup

cartridge



single-shot sandwich immunoassay, incubate and read once

wearable

continuous glucose monitors, 10–15 days [27]  
wound patch [28]; aptamer sensor, one week in vivo [29]



implant

Eversense implanted glucose monitor, 180–365 days [27]



1 min

1 h

1 day

1 week

1 month

1 year

time the electrode spends in the sample

EIS gives  
Firmware

each site's  $R_{ct} \parallel C_{dl}$  arc, trended over the electrode's life, from the same channel  
re-ranges, recalibrates, or retires the site

**Drift is systematic: averaging removes noise, not bias [25].**

# How impedance is measured: the same loop, no extra hardware

Castalia

Maxx Seminario

Target application

The chip

Analog front end

Operating modes

Analog blocks

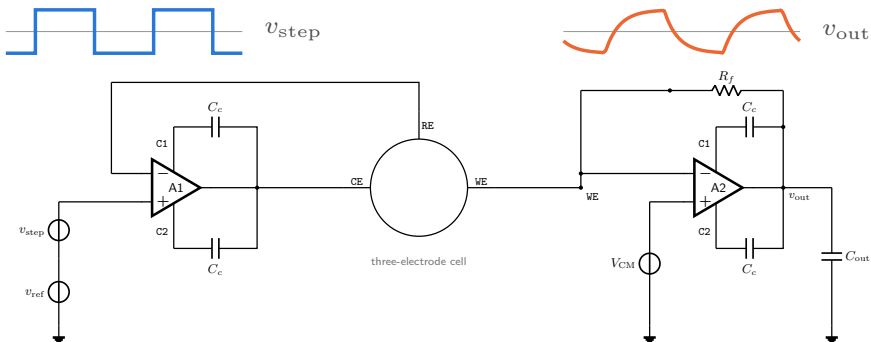
Digital

Vision

From currents to a score

References

Backup



Excite  
Sense  
Digitise  
Solve

DAC: a small two-code square about the DC setpoint  $v_{ref}$   
 $v_P$  across the cell;  $i_{WE} = -v_{out}/R_f$   
the SAR records both; firmware I/Q demodulation  
 $\hat{Z} = -R_f \hat{V}_P / \hat{V}_{OUT}$ , or a ratio to  $Z_{cal}$

# Magnitude and phase from codes: I/Q demodulation

Castalia

Maxx Seminario

Target application

The chip

Analog front end

Operating modes

Analog blocks

Digital

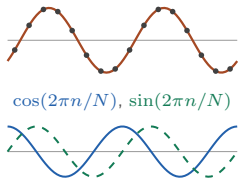
Vision

From currents to a score

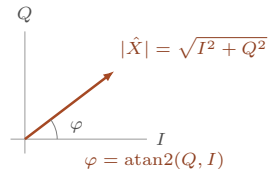
References

Backup

record  $c_n$ ,  $N$  per period,  $M$  periods



$$I = \frac{2}{MN} \sum c_n \cos \frac{2\pi n}{N}$$
$$Q = \frac{2}{MN} \sum c_n \sin \frac{2\pi n}{N}$$



Records  $v_P$  and  $v_{out}$   
Divide  $\hat{Z} = -R_f \hat{V}_P / \hat{V}_{OUT}$   
Ratio against  $Z_{cal}$ :  $R_f$  drops out

# The four loads behind the EIS plots

Castalia

Maxx Seminario

Target application

The chip

Analog front end

Operating modes

Analog blocks

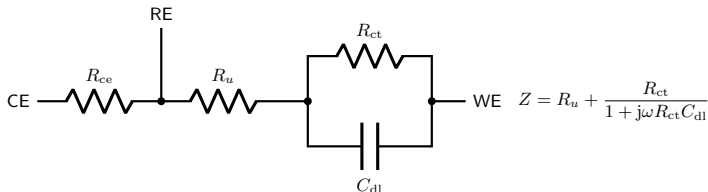
Digital

Vision

From currents to a score

References

Backup



| Load     | $R_{ct}$       | $C_{dl}$   | Gain code | Stands for                                                                                            |
|----------|----------------|------------|-----------|-------------------------------------------------------------------------------------------------------|
| resistor | –              | –          | $N = 0$   | an 11 k $\Omega$ resistor in place of the cell: no interface, so the trace is the channel's own error |
| fouled   | 10 k $\Omega$  | 10 $\mu$ F | $N = 0$   | a degraded electrode: adsorbed protein and biofilm open leakage paths and enlarge the interface       |
| typical  | 100 k $\Omega$ | 1 $\mu$ F  | $N = 20$  | an electrode in use                                                                                   |
| fresh    | 1 M $\Omega$   | 100 nF     | $N = 63$  | a new, clean surface                                                                                  |

$R_u = R_{ce} = 1 \text{ k}\Omega$  throughout. Fouling:  $R_{ct}$  falls and  $C_{dl}$  rises through an electrode's life; tracking that drift is what EIS is for here.

# Impedance spectroscopy on the same channel

Castalia

Maxx Seminario

Target application

The chip

Analog front end

Operating modes

Analog blocks

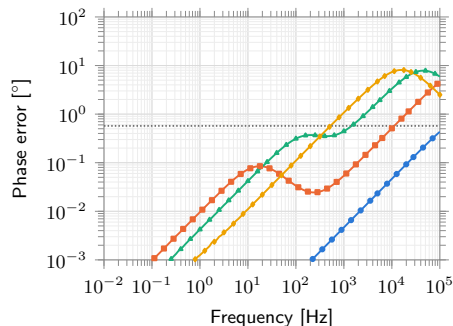
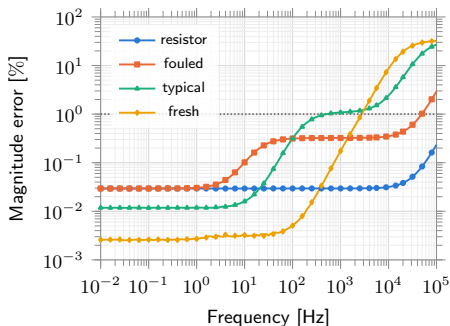
Digital

Vision

From currents to a score

References

Backup



Accuracy  $\pm 1\%$ ,  $\pm 0.6^\circ$  to 5 kHz;  $\pm 5\%$  to 27 kHz. Limit: A1's loop bandwidth  
 Range 246  $\Omega$  to 12.1 M $\Omega$ . Limits: TIA compliance below, one LSB of output above; 320  $R_f$  codes in between

The excitation amplitude never enters:  $Z$  is the ratio of two records through one converter, so DAC gain and droop cancel.

# What sets the EIS error, and what averaging buys

Castalia

Maxx Seminario

Target application

The chip

Analog front end

Operating modes

Analog blocks

Digital

Vision

From currents to a score

References

Backup

$$\varepsilon_{|Z|} \approx \underbrace{\frac{R_f/|Z|}{A_{\text{eff}}}}_{\text{systematic}} + \underbrace{\frac{i_n |Z|}{V_1} \sqrt{\frac{2}{NM}}}_{\text{random}}$$

$R_f/|Z|$  feedback resistor over the load: the gain code chosen

$A_{\text{eff}}$  effective loop gain,  $\approx 76$  dB measured (0.66 % at  $R_f/R = 40$ );  $A_{\text{OL}} \approx 83$  dB

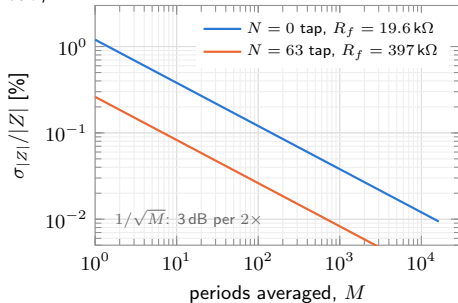
$i_n$  front-end current noise per sample: 273 nA at  $N=0$ , 61 nA at  $N=63$

$V_1$  excitation fundamental,  $\frac{4}{\pi} \frac{A}{2}$

$N, M$  samples per period, periods averaged

The systematic term is fixed for a gain code and load, and the  $Z_{\text{cal}}$  ratio removes it. The random term is what remains, and it falls as  $1/\sqrt{M}$ .

Random term against periods averaged, 11 k $\Omega$  load, 1 kHz



# Calibrating impedance

Castalia

Maxx Seminario

Target  
application

The chip

Analog front end

Operating modes

Analog blocks

Digital

Vision

From currents to  
a score

References

Backup

$$\hat{Z} = Z_{\text{cal}} \frac{(\hat{V}_P / \hat{V}_{\text{OUT}})}{(\hat{V}_P / \hat{V}_{\text{OUT}})_{\text{cal}}} \quad \text{same gain code, same frequency}$$

|          | On-chip poly, per decade                           | Off-chip standard, at test |
|----------|----------------------------------------------------|----------------------------|
| Cancels  | $R_f$ , its pole, DAC, LSB, loop-gain floor        | the same                   |
| Known to | $\sim 20\%$ chip to chip                           | $< 0.1\%$                  |
| Run      | any time, in the field                             | once, stored in flash      |
| Good for | trends: fouling is a change in one electrode's $Z$ | absolute ohms              |

**The poly need not be accurate, only stable. One stored number pins the scale; the on-chip ratio carries it through temperature and time.**

# The class-AB amplifier: one cell, both roles

Castalia

Maxx Seminario

Target application

The chip

Analog front end

Operating modes

Analog blocks

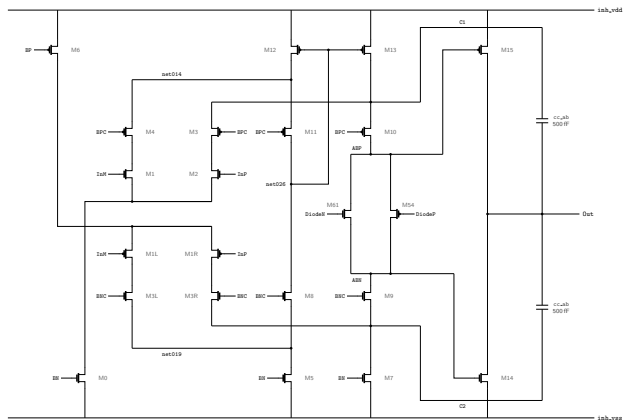
Digital

Vision

From currents to a score

References

Backup



complementary  
input pairs and  
their cascodes

diode-connected  
summing branch

class-AB  
drive

output

testbench  
compensation

# The amplifier across corners, with mismatch $\sigma$

Castalia

Maxx Seminario

Target  
application

The chip

Analog front end

Operating modes

Analog blocks

Digital

Vision

From currents to  
a score

References

Backup

| Parameter              | Unit | tt    | ff    | fs    | sf    | ss    | $\sigma$ (MC) |
|------------------------|------|-------|-------|-------|-------|-------|---------------|
| Open-loop gain         | dB   | 82.77 | 79.88 | 83.14 | 83.37 | 84.94 | 1.06          |
| Phase margin           | deg  | 73.52 | 74.70 | 74.04 | 72.86 | 73.34 | 1.42          |
| Dominant pole          | kHz  | 1.592 | 2.286 | 1.538 | 1.494 | 1.225 | 0.276         |
| Gain-bandwidth product | MHz  | 22.78 | 23.19 | 22.89 | 22.94 | 22.59 | 3.57          |
| Input offset voltage   | mV   | 0.622 | 0.649 | 0.624 | 0.569 | 0.456 | 5.201         |

# Bipolar accuracy through zero

Castalia

Maxx Seminario

Target application

The chip

Analog front end

Operating modes

Analog blocks

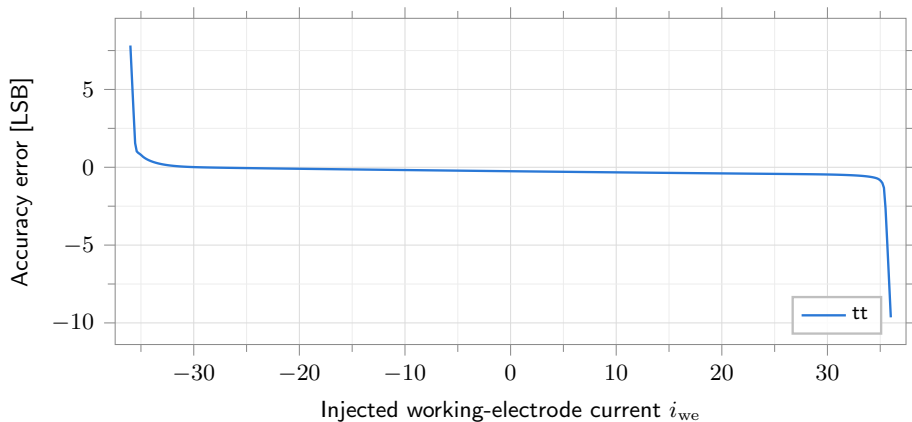
Digital

Vision

From currents to a score

References

Backup



# Programmable feedback resistor: one string, 320 codes

Castalia

Maxx Seminario

Target application

The chip

Analog front end

Operating modes

Analog blocks

Digital

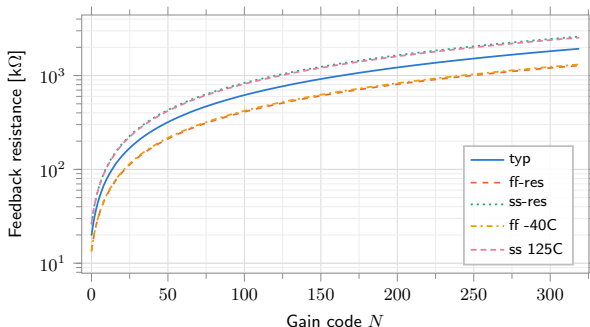
Vision

From currents to a score

References

Backup

- $R_f = 18 \text{ k}\Omega + 6 \text{ k}\Omega \cdot N$ ; full scale =  $0.80 \text{ V}/R_f$ , LSB =  $1.59 \text{ mV}/R_f$
- **Max  $\pm 40.7 \mu\text{A}$ , min full scale  $\pm 0.41 \mu\text{A}$ , resolution  $0.82 \text{ nA}$**
- Noise floor  $i_n^2 = e_n^2 (1 + R_f/|Z_{el}|)^2/R_f^2 + 4kT/R_f$ : 341 pA alone, 3.6–21 nA into an electrode (0.1–100 Hz, 560 k $\Omega$ )



| $N$ | $R_f$           | Full scale             | LSB     |
|-----|-----------------|------------------------|---------|
| 0   | 19.6 k $\Omega$ | $\pm 40.7 \mu\text{A}$ | 81 nA   |
| 63  | 397 k $\Omega$  | $\pm 2.02 \mu\text{A}$ | 4.0 nA  |
| 319 | 1.93 M $\Omega$ | $\pm 0.41 \mu\text{A}$ | 0.82 nA |

| Mismatch $\sigma$        | $\sigma/\text{typ}$ |        |
|--------------------------|---------------------|--------|
| $R(N=0)$                 | 71 $\Omega$         | 0.36 % |
| $R(N=319)$               | 0.7 k $\Omega$      | 0.04 % |
| Seam 63 $\rightarrow$ 64 | 504 $\Omega$        | 6.8 %  |

# 10-bit SAR ADC, extracted netlist

Castalia

Maxx Seminario

Target  
application

The chip

Analog front end

Operating modes

Analog blocks

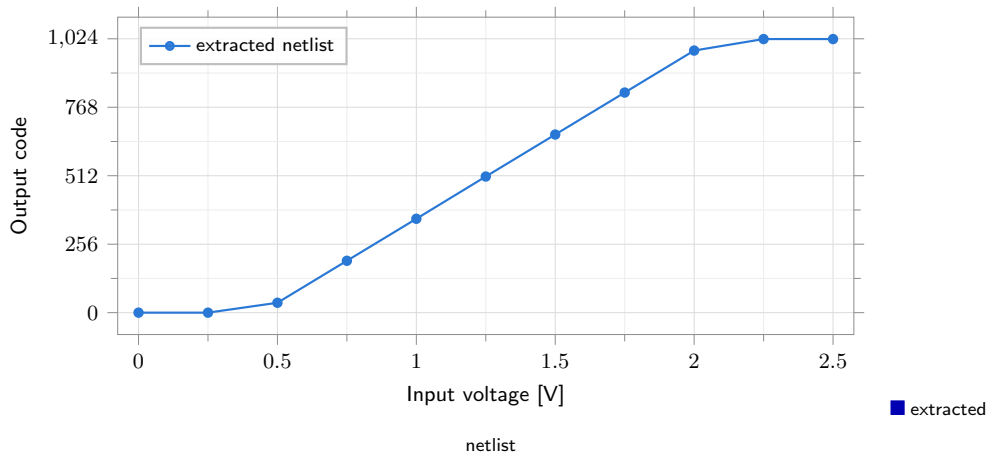
Digital

Vision

From currents to  
a score

References

Backup



**23 clocks per conversion, 870 kSa/s, 50 pJ. The capacitor array exists only in layout, so this is the one block characterised extracted.**

# SAR ADC across corners

Castalia

Maxx Seminario

Target application

The chip

Analog front end

Operating modes

Analog blocks

Digital

Vision

From currents to a score

References

Backup

| Parameter             | Unit    | tt     | ff     | fs     | sf     | ss     |
|-----------------------|---------|--------|--------|--------|--------|--------|
| Fitted slope          | codes/V | 628.59 | 628.59 | 628.40 | 628.73 | 628.13 |
| Effective LSB         | mV      | 1.5909 | 1.5909 | 1.5914 | 1.5905 | 1.5920 |
| Input at code 0       | V       | 0.4408 | 0.4408 | 0.4407 | 0.4408 | 0.4399 |
| Usable span           | V       | 1.6274 | 1.6274 | 1.6280 | 1.6271 | 1.6286 |
| Worst residual        | LSB     | 0.495  | 0.495  | 0.490  | 0.477  | 0.484  |
| RMS residual          | LSB     | 0.294  | 0.294  | 0.277  | 0.288  | 0.302  |
| Analog supply current | $\mu$ A | 6.36   | 4.79   | 5.45   | 8.44   | 15.43  |
| Energy per conversion | pJ      | 20.8   | 19.3   | 18.1   | 27.3   | 49.5   |

# Reference DAC across corners, with mismatch $\sigma$

Castalia

Maxx Seminario

Target  
application

The chip

Analog front end

Operating modes

Analog blocks

Digital

Vision

From currents to  
a score

References

Backup

| Parameter          | Unit          | tt       | ff       | fs       | sf       | ss       | $\sigma$ (MC) |
|--------------------|---------------|----------|----------|----------|----------|----------|---------------|
| Full-scale output  | V             | 2.499388 | 2.499388 | 2.499388 | 2.499388 | 2.499388 | 0.000001      |
| LSB size           | $\mu\text{V}$ | 610.351  | 610.351  | 610.351  | 610.351  | 610.351  | 0.0003        |
| INL, most positive | LSB           | 0.113    | 0.101    | 0.100    | 0.127    | 0.130    | 0.610         |
| INL, most negative | LSB           | -0.376   | -0.338   | -0.390   | -0.362   | -0.414   | 0.620         |
| DNL, most positive | LSB           | 0.002    | 0.001    | 0.002    | 0.002    | 0.002    | 1.128         |
| DNL, most negative | LSB           | -0.423   | -0.374   | -0.424   | -0.424   | -0.479   | 1.415         |

# The front end, against the field

Castalia

Maxx Seminario

Target application

The chip

Analog front end

Operating modes

Analog blocks

Digital

Vision

From currents to a score

References

Backup

|                                | Castalia (this work)         | Myshkin [9]       | Chen '19 [10] | Ghoreishizadeh '17 [11] | Lin '23 [12] |
|--------------------------------|------------------------------|-------------------|---------------|-------------------------|--------------|
| Modality                       | <b>CA/CV/DPV/EIS</b>         | CA/CV             | CA/CV         | CA/CV                   | CV/EIS       |
| Channels                       | <b>4, simultaneous</b>       | 1                 | 1             | 1                       | 1            |
| Current direction              | bipolar                      | unipolar          | bipolar       | bipolar                 | bipolar      |
| $I_{\max}$ ( $\mu\text{A}$ )   | $\pm 40.7$                   | +20               | $\pm 5$       | $\pm 20$                | $-48/+16$    |
| Resolution ( $\text{pA rms}$ ) | $341^a$                      | 56.5              | 41            | 0.47                    | 1.7          |
| ADC step ( $\text{pA}$ )       | $820^b$                      | 153               | n/a           | n/a                     | n/a          |
| Converter                      | 10-bit SAR per channel       | 12-bit dual-slope | TDC           | ADC                     | –            |
| EIS                            | firmware, $\pm 1\%$ to 5 kHz | –                 | –             | –                       | yes          |
| Power (mW)                     | $3.68^c$                     | 3.0               | 0.016         | 9.3                     | 1.0          |
| Process (nm)                   | <b>65</b>                    | 65                | 180           | 350                     | 180          |
| On-chip processor              | <b>5 RISC-V harts</b>        | 1 RISC-V hart     | none          | none                    | none         |
| On-chip ML                     | <b>NPU, 4 modes</b>          | NPU, 1 mode       | none          | none                    | none         |

<sup>a</sup>Input-referred noise, 0.1–100 Hz, ideal  $R_f$ , no electrode.

<sup>b</sup>One converter code at the maximum-gain tap; 81 nA at the lowest.

<sup>c</sup>Low-power mode: hart 0 awake, tiles gated, AFE off. One channel measuring adds 1.76 mW (tile) + 1.32 mW (AFE).

# Analog power: one channel

Castalia

Maxx Seminario

Target  
application

The chip

Analog front end

Operating modes

Analog blocks

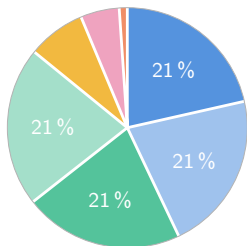
Digital

Vision

From currents to  
a score

References

Backup



- A1, control amplifier, 244  $\mu\text{W}$ , 21 %
- A2, transimpedance amplifier, 244  $\mu\text{W}$ , 21 %
- ADC input buffer, 244  $\mu\text{W}$ , 21 %
- debug buffer, 244  $\mu\text{W}$ , 21 %
- two R-2R ladders, 89  $\mu\text{W}$ , 8 %
- local bias generator, 59  $\mu\text{W}$ , 5 %
- SAR converter, 12  $\mu\text{W}$ , 1 %

## Power

|               |         |
|---------------|---------|
| One channel   | 1.32 mW |
| Four channels | 5.3 mW  |

# Castalia at a glance

## Castalia

### Maxx Seminario

## Digital

|                 |                                        |
|-----------------|----------------------------------------|
| Harts           | 5 (1 orchestrator, 4 channel tiles)    |
| ISA             | rv32imac_zba_zbb_zbs_zbc               |
| Clock           | 24 MHz rated, closed at 25 MHz         |
| Measured CPI    | 1.260 aggregate, nine benchmarks       |
| Private TCM     | 8 KiB per hart, 1-cycle                |
| Shared RAM      | 80 KiB behind a round-robin arbiter    |
| Synchronisation | 16 hardware mutexes, LR/SC, AMOs       |
| Interrupts      | claim/complete, any vector to any hart |
| Accelerator     | NPU: MLP, conv-1D, XNOR, GEMM          |
| Power domains   | 4 MTCMOS, one per channel tile         |
| Debug           | JTAG, RISC-V Debug Module              |
| Field boot      | NFC ISO 14443A harvested boot          |

## Analog and physical

|                   |                                                                |
|-------------------|----------------------------------------------------------------|
| Measurement sites | 4 identical, simultaneous                                      |
| Topology          | bipolar grounded-WE potentiostat                               |
| Electrode pads    | 12 (CE/RE/WE $\times$ 4)                                       |
| Converter         | 10-bit SAR per channel, extracted                              |
| Reference DAC     | 12-bit R-2R per channel                                        |
| Feedback resistor | $18\text{ k}\Omega + 6\text{ k}\Omega \cdot N$ , 320 codes     |
| Full scale        | $\pm 40.7\text{ }\mu\text{A}$ to $\pm 0.41\text{ }\mu\text{A}$ |
| EIS               | firmware mode of the same channels                             |
| Analog supply     | $2.5\text{ V}$ , $v_{\text{cm}} = 1.25\text{ V}$               |
| Process           | 65 nm CMOS, 1P8M, HVT cells                                    |
| Die / package     | $3.0 \times 3.0\text{ mm}$ / LQFP-100                          |
| Digital power     | 12.8 mW core post-route, SS, 25 MHz                            |

Digital numbers from the hardened flow; analog numbers schematic-level, except the SAR, which is extracted.

# Digital, as closed

Castalia

Maxx Seminario

Target  
application

The chip

Analog front end

Operating modes

Analog blocks

Digital

Vision

From currents to  
a score

References

Backup

## Digital, post-route

|           |         |
|-----------|---------|
| Internal  | 8.69 mW |
| Switching | 9.99 mW |
| Leakage   | 4.70 mW |

|              |                |
|--------------|----------------|
| Core logic   | 12.8 mW        |
| Pad ring*    | 10.5 mW        |
| <b>Total</b> | <b>23.4 mW</b> |

## One channel tile

|       |          |
|-------|----------|
| Total | 1.761 mW |
| Gated | 0        |

## Measured CPI, timed kernels

| Benchmark | Instr.  | CPI   |
|-----------|---------|-------|
| spmv      | 802 447 | 1.143 |
| multiply  | 20 892  | 1.163 |
| vvadd     | 2 408   | 1.374 |
| median    | 4 015   | 1.398 |
| qsort     | 123 205 | 1.398 |
| rsort     | 146 548 | 1.412 |
| dhrystone | 200 523 | 1.504 |
| towers    | 3 747   | 1.682 |
| memcpy    | 11 017  | 1.726 |

|                  |                  |              |
|------------------|------------------|--------------|
| <b>Aggregate</b> | <b>1 314 802</b> | <b>1.260</b> |
|------------------|------------------|--------------|

SS 0.9 V 125 °C, 25 MHz, statistical activity 0.2; d13d cut. \*Pads at the statistical activity, to be optimised.

■ hardened flow, post-route

# Power gating: one switched rail per channel tile

Castalia

Maxx Seminario

Target application

The chip

Analog front end

Operating modes

Analog blocks

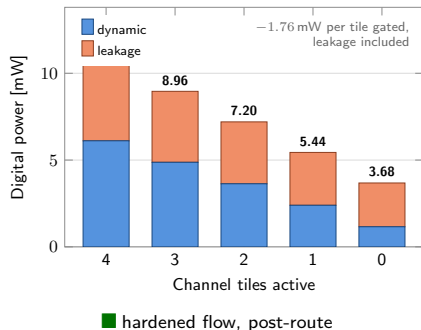
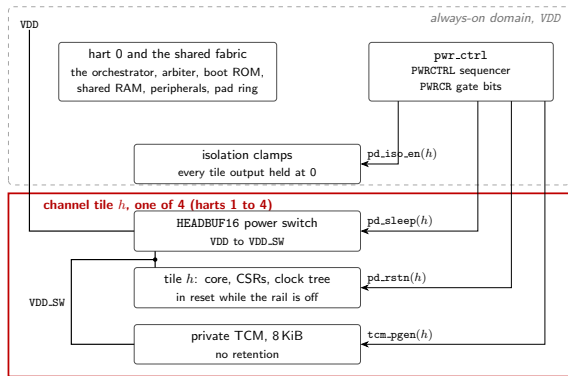
Digital

Vision

From currents to a score

References

Backup



**Cold gating:** the header cuts the rail, so a parked tile draws neither dynamic nor leakage power. Each tile gated removes 1.76 mW; the always-on band keeps 3.68 mW.

# Vision: the same channel, without the cartridge

Castalia

Maxx Seminario

Target application

The chip

Analog front end

Operating modes

Analog blocks

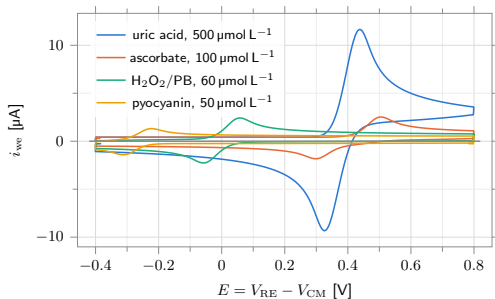
Digital

Vision

From currents to a score

References

Backup



---

## Already on the die

---

|         |                                             |
|---------|---------------------------------------------|
| 4 sites | concurrent, each its own DAC, SAR, hart     |
| EIS     | electrode state without a second instrument |
| NFC     | harvested boot, battery-free readout        |
| MTCMOS  | a parked site costs nothing                 |
| NPU     | the decision stays on the patient           |

---

Left: a small-molecule wound panel, uric acid, ascorbate,  $\text{H}_2\text{O}_2$  and pyocyanin, through the same channel.

The ovarian panel stays a serum test: CA-125 is an epitope on a 3–5 MDa mucin and does not reach interstitial fluid. The wearable is a different, reagentless panel on the same silicon; the implant is the same plus a fouling problem that EIS can watch but not solve.

# The fusion runs on the NPU

Castalia

Maxx Seminario

Target  
application

The chip

Analog front end

Operating modes

Analog blocks

Digital

Vision

From currents to  
a score

References

Backup

$$s = \sigma \left( \sum_{k=1}^4 w_k \hat{c}_k + b \right)$$

|         |                                                   |
|---------|---------------------------------------------------|
| Input   | peak current, peak potential, baseline $\times 4$ |
| Mode    | 0, dense layer, one sigmoid                       |
| Output  | one risk score                                    |
| Weights | trained off-chip, staged by firmware              |

## Per site

|           |                                   |
|-----------|-----------------------------------|
| Hart      | 1, ownership enforced in hardware |
| Potential | own 12-bit DAC                    |
| Converter | own 10-bit SAR                    |
| Power     | own MTCMOS domain                 |

The logistic form of the cleared two-marker algorithm, extended to four inputs and moved from the central laboratory onto the electrode.

# References (1/5)

Castalia

Maxx Seminario

Target application

The chip

Analogue front end

Operating modes

Analogue blocks

Digital

Vision

From currents to a score

References

Backup

- [1] NCI, *SEER Cancer Stat Facts*, per-site pages, [seer.cancer.gov/statfacts](https://seer.cancer.gov/statfacts), accessed 20 Sep. 2026. Stage, survival: SEER 21, 2016–22; cases, deaths: 2026 est.
- [2] D. C. Grossman *et al.*, “Screening for ovarian cancer: USPSTF recommendation statement,” *JAMA* 319:588, 2018.
- [3] USPSTF, *A and B Recommendations; Prostate Cancer: Screening*, 2018; *Pancreatic Cancer: Screening*, 2019. [uspreventiveservicestaskforce.org](https://uspreventiveservicestaskforce.org).
- [4] U. Menon *et al.*, “Ovarian cancer population screening and mortality after long-term follow-up in UKCTOCS,” *Lancet* 397:2182, 2021.
- [5] F. Chen *et al.*, “Clinical analysis of four serum tumor markers in 458 patients with ovarian tumors: combined use of HE4, CA125, CA19-9, and CEA,” *Cancer Manag. Res.* 10:1313, 2018.
- [6] R. G. Moore *et al.*, “A novel multiple marker bioassay utilizing HE4 and CA125 for the prediction of ovarian cancer in patients with a pelvic mass,” *Gynecol. Oncol.* 112:40, 2009.

## References (2/5)

Castalia

Maxx Seminario

Target application

The chip

Analog front end

Operating modes

Analog blocks

Digital

Vision

From currents to a score

References

Backup

- [7] USPSTF, *Grade Definitions* (current, after July 2012), [uspreventiveservicestaskforce.org](https://uspreventiveservicestaskforce.org), accessed 20 Sep. 2026.
- [8] P. Charkhchi *et al.*, "CA125 and ovarian cancer: a comprehensive review," *Cancers* 12:3730, 2020.
- [9] M. A. Seminario, S. McRobert, A. Uerling, P. Aberson, S. Balkır, J. A. Schmitz and E. J. Markvicka, "An electrochemical sensing SoC for autonomous wound monitoring," *IEEE ISCAS*, 2026.
- [10] Y.-C. Chen, S.-Y. Lu and Y.-T. Liao, "A microwatt dual-mode electrochemical sensing current readout with current-reducer ramp waveform generation," *IEEE TBioCAS* 13(6):1163, 2019.
- [11] S. S. Ghoreishizadeh *et al.*, "A differential electrochemical readout ASIC with heterogeneous integration of bio-nano sensors for amperometric sensing," *IEEE TBioCAS* 11(5):1148, 2017.
- [12] Y.-J. Lin *et al.*, "A multimodality electrochemical and impedance spectroscopy system-on-a-chip with temperature sensing and impedance-boosting techniques," *IEEE TBioCAS* 17(4):857, 2023.

## References (3/5)

Castalia

Maxx Seminario

Target  
application

The chip

Analogue front end

Operating modes

Analogue blocks

Digital

Vision

From currents to  
a score

References

Backup

- [13] A. J. Bard and L. R. Faulkner, *Electrochemical Methods: Fundamentals and Applications*, 2nd ed., Wiley, 2001 (Eqs. 3.3.11, 4.4.2, 4.4.29, Sec. 1.4.2, Eq. 6.2.19).
- [14] M. S. Wilson and W. Nie, "Multiplex measurement of seven tumor markers using an electrochemical protein chip," *Anal. Chem.*, 2006.
- [15] A. Kovářová *et al.*, "Biosensor with electrochemically active nanocomposites for signal amplification and simultaneous detection of three ovarian cancer biomarkers," *Electrochim. Acta*, 2023.
- [16] V. Mirčeski *et al.*, "Square-wave voltammetry: a review on the recent progress," *Electroanalysis*, 2013.
- [17] D. Britz and J. Strutwolf, *Digital Simulation in Electrochemistry*, 4th ed., Springer, 2016 (Ch. 3 box method, Ch. 7 expanding grid).
- [18] S. W. Feldberg, "Digital simulation: a general method for solving electrochemical diffusion-kinetic problems," *Electroanal. Chem.* 3:199, 1969; J. E. B. Randles, *Discuss. Faraday Soc.* 1:11, 1947.

## References (4/5)

- [19] Jog *et al.*, Verilog-A three-electrode model with kinetics and mass transport for co-simulation with interface circuits, *IEEE SENSORS*, 2024, doi:10.1109/SENSORS60989.2024.10784491; S. Madec *et al.*, *IEEE TCAS-I* 66(6):2188, 2019.
- [20] P. Fanjul-Bolado, M. B. González-García and A. Costa-García, "Amperometric detection in TMB/HRP-based assays," *Anal. Bioanal. Chem.* 382:297, 2005.
- [21] M. Bilgi Kamaç *et al.*, "Point-of-care testing: a disposable label-free electrochemical CA125 and HE4 immunosensors for early detection of ovarian cancer," *Biomed. Microdevices* 25:18, 2023.
- [22] D. Kang, F. Ricci, R. J. White and K. W. Plaxco, "Survey of redox-active moieties for application in multiplexed electrochemical biosensors," *Anal. Chem.* 88(21):10452, 2016.
- [23] V. Clark, M. Aller Pellitero and N. Arroyo-Currás, "Explaining the decay of nucleic acid-based sensors under continuous voltammetric interrogation," *Anal. Chem.* 95(11):4974, 2023.
- [24] K. K. Leung *et al.*, "Elucidating the mechanisms underlying the signal drift of electrochemical aptamer-based sensors in whole blood," *ACS Sens.* 6:3340, 2021.

# References (5/5)

Castalia

Maxx Seminario

Target application

The chip

Analogue front end

Operating modes

Analogue blocks

Digital

Vision

From currents to a score

References

Backup

- [25] H. Li *et al.*, "Dual-reporter drift correction to enhance the performance of electrochemical aptamer-based sensors in whole blood," *J. Am. Chem. Soc.* 138:15809, 2016.
- [26] J. Sabatè del Río *et al.*, "An antifouling coating that enables affinity-based electrochemical biosensing in complex biological fluids," *Nat. Nanotechnol.* 14:1143, 2019.
- [27] Manufacturer labelling and FDA clearances: Dexcom G7 (10-day; 15-day, adults), Abbott FreeStyle Libre 3 (14 days), Senseonics Eversense 365 (implanted, 365 days; FDA clearance 17 Sep. 2024).
- [28] E. Shirzaei Sani *et al.*, "A stretchable wireless wearable bioelectronic system for multiplexed monitoring and combination treatment of infected chronic wounds," *Sci. Adv.* 9(12):eadf7388, 2023.
- [29] Son *et al.*, "Continuous, week-long, seconds-resolved in vivo drug measurements performed with a xenonucleic acid-employing electrochemical, aptamer-based sensor," *J. Am. Chem. Soc.* 148(9):10016, 2026.

## Backup: electrode model validation

Castalia

Maxx Seminario

Target application

The chip

Analog front end

Operating modes

Analog blocks

Digital

Vision

From currents to a score

References

Backup

| Scan rate<br>$\text{mV s}^{-1}$ | $i_p$ Randles–Ševčík<br>$\mu\text{A}$ | $i_p$ model<br>$\mu\text{A}$ | Error<br>% | Error, $C_{\text{dl}} \rightarrow 0$<br>% | $\Delta E_p$<br>$\text{mV}$ |
|---------------------------------|---------------------------------------|------------------------------|------------|-------------------------------------------|-----------------------------|
| 25                              | 0.951                                 | 0.952                        | +0.08      | −0.45                                     | 58.78                       |
| 50                              | 1.345                                 | 1.349                        | +0.32      | −0.42                                     | 59.85                       |
| 100                             | 1.902                                 | 1.915                        | +0.67      | −0.38                                     | 60.41                       |
| 200                             | 2.690                                 | 2.721                        | +1.16      | −0.33                                     | 62.33                       |
| 400                             | 3.804                                 | 3.875                        | +1.86      | −0.25                                     | 64.74                       |

Peak current against Randles–Ševčík and  $\Delta E_p$  against Nicholson [13], reversible couple,  $C_{\text{dl}} = 0$  column isolates the faradaic model.

## Backup: USPSTF screening grades

Castalia

Maxx Seminario

Target application

The chip

Analog front end

Operating modes

Analog blocks

Digital

Vision

From currents to a score

References

Backup

| Grade    | Certainty of net benefit          | Definition [7]                                                      | Cancer screening [2, 3]                                                                        |
|----------|-----------------------------------|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| <b>A</b> | high, substantial                 | Recommends the service. Offer or provide it.                        | cervix (2018), colorectal 50–75 (2021)                                                         |
| <b>B</b> | high moderate, or moderate        | Recommends the service. Offer or provide it.                        | breast 40–74 (2024), lung LDCT 50–80 (2021), colorectal 45–49 (2021) prostate PSA 55–69 (2018) |
| <b>C</b> | at least moderate, small          | Offer selectively, by professional judgment and patient preference. |                                                                                                |
| <b>D</b> | no net benefit, or harms outweigh | Recommends <i>against</i> the service. Discourage its use.          | <b>ovary (2018)</b> , pancreas (2019)                                                          |
| <b>I</b> | cannot be determined              | Evidence insufficient, poor, or conflicting.                        | —                                                                                              |
| none     | —                                 | No USPSTF statement exists for the condition.                       | uterus                                                                                         |

Ovarian D is a verdict on a test that was tried, CA-125 plus transvaginal ultrasound [2, 4], not on the absence of one.